Descriptive statistics

Lecture and worked examples

Maths Explorer

Frequencies, relative frequencies and mean

A data set describes a characteristic over a population. If the value x_i appears n_i times, n_i is its frequency count. The nonzero total count is $N = \sum_{i=1}^{p} n_i$; its relative frequency is $f_i = n_i/N$. The mean is

$$\overline{x} = \frac{\sum_{i=1}^{p} n_i x_i}{N} = \sum_{i=1}^{p} f_i x_i.$$

The range is $x_{\max} - x_{\min}$.

Frequencies, relative frequencies and mean

We study the shoe sizes of 30 students:

36	42	40	41	41	38	39	43	42	42	42	40	36	40	39
41	40	38	44	42	40	42	43	38	43	36	41	41	44	43

Complete the counts, calculate the relative frequencies, the mean and the range.

Worked answer: Frequencies, relative frequencies and mean

Counts: $3, 0, 3, 2, 5, 5, 6, 4, 2$, with sum 30. The relative frequencies are these counts divided by 30, that is as percentages 10; 0; 10; about 6.67; 16.67; 16.67; 20; 13.33; 6.67. $\sum n_i x_i = 1217$, so $\overline{x} = 1217/30 \approx 40.57$. The range is 8. The mean is not necessarily a value of the data set. The sum of the rounded percentages may differ slightly from 100.

Median and cumulative counts

A median is a number such that at least half of the values are less than or equal to it, and at least half are greater than or equal to it. In the ordered data set of N values, if N is odd, take the value of rank $(N+1)/2$; if N is even, choose the mean of the values of ranks $N/2$ and $N/2+1$.

Median and cumulative counts

With repeated values, more than 50% of the values can be less than or equal to the median.

Median and cumulative counts

Complete the ascending cumulative counts (ACC) of the shoe sizes and determine the median.

Worked answer: Median and cumulative counts

The central ranks are 15 and 16. The ACC goes from 13 for 40 to 18 for 41: both central values equal 41, so $M_e = 41$. At least half of the sizes are less than or equal to 41, and at least half are greater than or equal to 41.

Median and cumulative counts

Calculate the medians of $1; 2; 2; 8; 9$ and of $1; 2; 8; 9$.

Worked answer: Median and cumulative counts

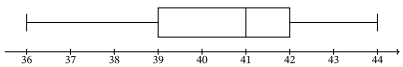
The third value of the first data set is 2. In the second, $(2 + 8)/2 = 5$; the chosen median is not a value of the data set.

Quartiles and box plot

The first quartile Q_1 is the value of rank $\lceil N/4 \rceil$: the smallest value of the data set such that at least 25% of the values are less than or equal to it. Q_3 is the value of rank $\lceil 3N/4 \rceil$. The interquartile range is $Q_3 - Q_1$. The box plot shows the minimum, Q_1, the median, Q_3 and the maximum.

Quartiles and box plot

Determine Q_1, Q_3 and the interquartile range of the shoe sizes, then draw the box plot.



Worked answer: Quartiles and box plot

$N/4 = 7.5$: rank 8, so $Q_1 = 39$. $3N/4 = 22.5$: rank 23, so $Q_3 = 42$. Interquartile range 3. The plot uses $(36; 39; 41; 42; 44)$.

Quartiles and box plot

Determine the quartiles of 1; 2; 2; 3; 4; 4; 4; 5. Are exactly 25% of the values less than or equal to Q_1?

Worked answer: Quartiles and box plot

$Q_1 = 2$ (rank 2), $Q_3 = 4$ (rank 6). Here $3/8 = 37.5\%$ of the values are less than or equal to Q_1. One must say “at least”.

Adding or removing a value

Adding a to a data set of size N gives $\overline{x}' = (N\overline{x} + a)/(N + 1)$. Removing one occurrence of a, for $N > 1$, gives $\overline{x}' = (N\overline{x} - a)/(N - 1)$. Use the exact sum, with no intermediate rounding. Re-examine the median in the new ordered data set.

Adding or removing a value

In the shoe-size data, the sum is 1217 and the median 41.

1. Add a size 36 and calculate the new statistics.
2. Starting again from the initial data, remove a size 44 and repeat.

Worked answer: Adding or removing a value

1. Sum 1253, count 31: $\overline{x}' = 1253/31 \approx 40.42$. The central rank 16 still gives 41. 2. Sum 1173, count 29: $\overline{x}'' = 1173/29 \approx 40.45$. The central rank 15 gives 41. The mean decreases in both cases; the median is unchanged. The situations are independent.

Adding or removing a value

Justify that adding the mean does not change the mean. Can the range decrease when a value is added?

Worked answer: Adding or removing a value

If $a = \overline{x}$, $(N\overline{x} + \overline{x})/(N + 1) = \overline{x}$. When adding, the minimum can only stay the same or decrease, the maximum stay the same or increase: the range does not decrease. When removing, it does not increase.

Variance and standard deviation

$V = \dfrac{\sum n_i (x_i - \overline{x})^2}{N}$ is the variance; $\sigma = \sqrt{V}$ is the standard deviation. It measures the spread around the mean and has the same unit as the data. One can also compute $V = \dfrac{\sum n_i x_i^2}{N} - \overline{x}^2$.

Variance and standard deviation

Calculate the variance and the standard deviation of the shoe sizes.

Worked answer: Variance and standard deviation

$\sum n_i x_i^2 = 49519$, so

$$V = \frac{49519}{30} - \left(\frac{1217}{30}\right)^2 = \frac{4481}{900} \approx 4.979.$$

$\sigma = \sqrt{4481}/30 \approx 2.23$. The standard deviation is not the maximum distance from the mean.

Variance and standard deviation

Compare A : $8; 10; 12$ and B : $0; 10; 20$ using the mean-standard deviation pair and the median-interquartile range pair.

Worked answer: Variance and standard deviation

The means and medians are all 10. $V_A = 8/3$, $\sigma_A \approx 1.63$; $V_B = 200/3$, $\sigma_B \approx 8.16$. Quartiles: $8, 12$ for A; $0, 20$ for B. Interquartile ranges 4 and 20. B is much more spread out despite identical centres.

Linearity of the mean

Replacing each x_i by $kx_i + c$ gives a mean $k\overline{x} + c$, a standard deviation $|k|\sigma$ and a variance k^2V. Adding a constant changes neither variance nor standard deviation.

Linearity of the mean

Initial mean 12/20, standard deviation 3. Calculate the new statistics for three independent modifications, with no capping of marks: add 1.5; multiply by 0.8; multiply by 1.1 then add 0.5.

Worked answer: Linearity of the mean

Means: 13.5, 9.6, 13.7. Standard deviations: 3, 2.4, 3.3. Multiplying the marks by 1.1 raises them; this does not mean making the assessment harder. With a cap at 20, the transformation would no longer be affine on all the data.

Python

```
from math import sqrt

def moyenne_ecart_type(valeurs):
    n = len(valeurs)
    if n == 0:
        raise ValueError("liste vide")
    m = sum(valeurs)/n
    v = sum((x-m)**2 for x in valeurs)/n
    return m, sqrt(v)
```

Linearity of the mean

Explain the program; predict its result for `[8, 10, 12]`.

Worked answer: Linearity of the mean

It checks that the list is nonempty, calculates the mean, the mean of the squared deviations, then its root. Result: $(10, \sqrt{8/3}) \approx (10, 1.633)$.

Continuous data grouped into classes

A continuous quantitative variable can take every value in an interval. Here, the observations are grouped into classes of equal width; the centre of $[a, b)$ is $(a + b)/2$. If the class means m_i are known, the exact overall mean is $\sum n_i m_i / N$. Otherwise, replacing them by the centres gives an estimate justified by a uniform model within each class.

Continuous data grouped into classes

Complete, estimate the mean from the centres and determine the median class.

Worked answer: Continuous data grouped into classes

Width 10 cm, total count 100. Relative frequencies as percentages: $8, 22, 35, 28, 7$. Centres: $155, 165, 175, 185, 195$ cm. The weighted sum of the centres is 17540, so the estimated mean is 175.4 cm. The central ranks $50, 51$ lie in $[170, 180)$, the median class.

Continuous data grouped into classes

If the exact class means were 154; 166; 174; 184; 196 cm, what would the overall mean be?

Worked answer: Continuous data grouped into classes

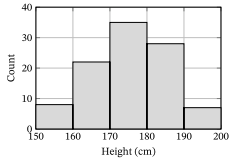
$(8 \times 154 + 22 \times 166 + 35 \times 174 + 28 \times 184 + 7 \times 196)/100 = 174.98$ cm.

Histogram and cumulative frequencies

The area of each rectangle of a histogram is proportional to the count. For equal widths, the heights can be the counts. The cumulative frequency polygon joins the points (upper bound; cumulative frequency), starting at the first lower bound with 0.

Histogram and cumulative frequencies

Construct both graphs for the heights of the previous example.



Histogram and cumulative frequencies

Under the assumption of uniformity in $[170, 180)$, the cumulative frequency rises linearly from 30% to 65%. Look for the abscissa corresponding to 50%.

Histogram and cumulative frequencies

Calculate this estimate and state its scope.

Worked answer: Histogram and cumulative frequencies

Solve $(m - 170)/10 = (50 - 30)/(65 - 30)$, giving $m = 170 + 10 \times 20/35 \approx 175.71$ cm. This is an estimate under an assumption of uniformity, not the exact median of the individual data. The class counts alone do not determine the latter.